3. Which of the following is not a component of downstream processing? (NEET-II 2016)
a) Separation b) Purification
c) Preservation **d) Expression**

4. Which of the following is not a feature of the plasmids? (NEET-I 2016)
a) Transferable **b) Single-stranded**
c) Independent replication
d) Circular structure

5. Which of the following is not required for nay of the techniques of DNA fingerprinting available at present? (NEET-I 2016)
a) Restriction enzymes
b) DNA-DNA hybridization
c) Polymerase chain reaction
d) Zinc finger analysis

6. Which vector can clone only a small fragment of DNA? (AIPMT 2014)
a) Bacterial artificial chromosome
b) Yeast artificial chromosome
c) Plasmid d) Cosmid

7. The colonies of recombinant bacteria appear white in contrast to blue colonies of non-recombinant bacteria because of (NEET 2013)
a) Insertional inactivation of alpha galactosidase in recombinant bacteria.
b) Inactivation of glycosidase enzyme in recombinant bacteria.
c) N**on-recombinant bacteria containing beta galactosidase.**
d) Insertional inactivation of alpha galactosidase in non-recombinant bacteria.

8. During the process of isolation of DNA, chilled ethanol is added to (Karnataka NEET 2013)
a) **Precipitate DNA**
b) Break open the cell to release DNA
c) Facilitate action of restriction enzymes
d) Remove proteins such as histones.

9. For transformation, micro-particles coated with DNA to be bombarded with gene gun are made up of (AIPMT 2012)
a) Silver or platinum b) Platinum or zinc
c) Silicon or platinum **d) Gold or tungsten.**

10. Biolistics (gene-gun) is suitable for (AIPMT Mains 2012)
a) disarming pathogen vectors
b) transformation of plant cells
c) constructing recombinant DNA by joining with vectors
d) DNA fingerprinting.

11. Genetic engineering is possible because (CBSE 1998)
a) phenomenon of transduction in bacteria understood
b) we can see DNA by electron microscope
c) we can cut DNA at specific sites by endonuclease like DNAase I
d) restriction endonuclease purified from bacteria can be used invitro

12. Genetic Engineering is (BHU 2003)
a) Making artificial genes
b) **Hybridisation of DNA of one organism to that of the others**
c) Production of alcohol by using microorganisms
d) Making artificial limbs, diagnostic instruments such as ECG, EFG, etc.

13. Ligase is used for (AMU 2006)
a) **Joining of two DNA fragments**
b) Separating DNA
c) DNA polymerase reaction
d) All of these

14. In genetic engineering, gene of interest is transferred to the host cell through a vector. Consider the following four agents (1-4) in this regard and select the correct option about which one or more of these can be used as vectors
1. A bacterium 2. Plasmid
3. Plasmodium 4. Bacteriophage

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